🌐 GLA UNIVERSITY // HIGH PERFORMANCE COMPUTING (HPC) & HETEROGENEOUS AI SYSTEMS

Master Curriculum Index, Bidirectional Knowledge Graph & Architectural Navigation Ledger

Course Code: GLA-HPC-BCSG701 (Block Teaching Delivery)
Host Institution: Department of Computer Engineering & Applications, GLA University, Mathura
Contracting Foundation: Edulateral Foundation, Bangalore
Industry Alliances: Intel Unnati Labs & NEC Supercomputing Center of Excellence
Instructional Split:

- **Lead HPC & AI Architecture Trainer (Expert Day 1 & Day 5)**: **Shivam Gupta (~~SEED~~)**
- **Co-Trainer & Lab Lead (Day 2, 3, 4 Co-Lead & Operations)**: **Rajat Garg (Rāmapriya Das)**
- **Target Audience**: 7th Semester B.Tech Computer Science & Engineering (HPC / AI Elective)

> 🚀 **Direct Presentation Hub**: All Day-wise PowerPoint decks (`.pptx`), speaker notes (`.md`), slide OCR transcripts, and syllabus guides are centralized in the top-level directory: `MAIN_PRESENTATIONS/` (with subfolders for Day 1 through Day 5).

🗺 1. Master System Knowledge Graph & AST Community Index

The workspace codebase and courseware have been analyzed, AST-extracted, and clustered using **Graphify Knowledge Engine** (298 nodes, 279 structural edges, 88 communities). Below is the bidirectional navigational ledger mapping every core concept to its physical implementation, presentation slide, interactive lab, and assessment asset.

```
graph TD
    subgraph S1["DAY 1: HPC FOUNDATIONS, LINUX & INTEL oneAPI"]
        D1_Theory["Silicon Microarchitecture, Cache Wall, AVX-512"]
        D1_Linux["Linux Kernel task_struct, Shell Scripting & SSH"]
        D1_Base["Intel oneAPI Base Toolkit (DPC++/SYCL, USM)"]
        D1_HPC["Intel oneAPI HPC Toolkit (icx, icpx, ifx, Intel MPI)"]
        D1_Lab["Hands-on Lab: Google Colab & Cluster DPC++ Kernels"]
        D1_MCQ["Day 1 MCQ Assessment Bank (25 Questions)"]
        D1_Theory --> D1_Linux --> D1_Base --> D1_HPC --> D1_Lab --> D1_MCQ
    end

    subgraph S2["DAY 2: PARALLEL COMPUTING, OPENMP, MPI & oneDAL"]
        D2_OpenMP["OpenMP Fork-Join & False Sharing (MESI Protocol)"]
        D2_MPI["MPI Point-to-Point & Collective Ring Topologies"]
        D2_oneDAL["Intel oneDAL & scikit-learn-intelex (30x Speedup)"]
        D2_Lab["Multi-Threaded & Multi-Node Cluster Benchmarking"]
        D2_OpenMP --> D2_MPI --> D2_oneDAL --> D2_Lab
    end

    subgraph S3["DAY 3: DEEP LEARNING ON HPC & SLURM"]
        D3_DL["TensorFlow-MKL, oneDNN GEMM & Intel AMX Systolic"]
        D3_Slurm["Slurm Workload Manager (slurmctld, slurmd, sbatch)"]
        D3_Horovod["Horovod Ring-AllReduce vs Parameter Server"]
        D3_Lab["Distributed DL Job Submission via Slurm Scripts"]
        D3_DL --> D3_Slurm --> D3_Horovod --> D3_Lab
    end

    subgraph S4["DAY 4: DISTRIBUTED AI & INTEL VTUNE PROFILING"]
        D4_VTune["Intel VTune Profiler (TMAM: Retiring vs Memory-Bound)"]
        D4_Roofline["Cache-Aware Roofline Model (Arithmetic Intensity)"]
        D4_NUMA["Dual-Socket NUMA Memory Latency & numactl"]
        D4_Lab["Memory Hotspot & Thread Concurrency Profiling Lab"]
        D4_VTune --> D4_Roofline --> D4_NUMA --> D4_Lab
    end

    subgraph S5["DAY 5: GENERATIVE AI, LLMS ON INTEL HPC & OPENVINO"]
        D5_GenAI["Silicon Physics of Transformers & Memory Bandwidth Wall"]
        D5_Hardware["Intel Xeon AMX vs Habana Gaudi 2/3 (RoCE v2 Networking)"]
        D5_OpenVINO["OpenVINO Toolkit: NNCF INT8/INT4 Weight-Only Quantization"]
        D5_FineTune["Distributed Fine-Tuning: IPEX, DeepSpeed & LoRA on HPC"]
        D5_Lab["OpenVINO LLM Compression Lab & FastAPI Microservice"]
        D5_Cap["Capstone Project Evaluation & Certification"]
        D5_GenAI --> D5_Hardware --> D5_OpenVINO --> D5_FineTune --> D5_Lab --> D5_Cap
    end

    S1 --> S2 --> S3 --> S4 --> S5
```

📂 2. Comprehensive 5-Day Curriculum Navigation Ledger

Day	Modules & Silicon Focus	Primary Slide Decks	Master Dossiers & Teaching Scripts	Hands-on Labs & Code Assets	Assessment & MCQs

Day 1	**HPC Fundamentals, Linux & Intel oneAPI**
	• Von Neumann Bottleneck & Memory Wall
	• AVX-512 SIMD & Cache Hierarchy (L1/L2/L3)
	• Linux Shell Scripting & Automated Audits
	• SSH Keygen & Multi-Hop ProxyJump
	• Intel oneAPI Base Toolkit (DPC++/SYCL, USM)
	• Intel HPC Toolkit (`icx` , `icpx` , `ifx` , Intel MPI)
		`Day_01_HPC_Foundations_Linux_oneAPI_LMS.pptx`
		`Day_01_High_Performance_AI_Foundations.pptx`
			`Day-01 Master Dossier`
			`Day-01 Line-by-Line Teaching Guide`
			`Slide & Lab Synchronization Playbook`
			`Day-01 Sync Demo Rehearsal Script`
				`Day_01_Master_Lab.ipynb`
				`live_demo/orchestrator.sh`
				`student_lab_practice/`
					`Day-01_MCQs.md`
					25 MCQs with Explanations

Day	Modules & Silicon Focus	Primary Slide Decks	Master Dossiers & Teaching Scripts	Hands-on Labs & Code Assets	Assessment & MCQs
Day 2	**Parallel Computing, OpenMP, MPI & oneDAL** • Amdahl's Law vs Gustafson's Law • OpenMP Multi-threading & False Sharing Trap • MESI Cache Coherence Bus Invalidation • MPI Point-to-Point & Collective Operations • Intel oneDAL Accelerated Scikit-Learn	`Day_02_Traditional_ML_Parallel_Computing_LMS.pptx`	`Day-02 Master Dossier`	`Day-02 live_demo/` `student_lab_practice/`	`Day-02_MCQs.md` 25 MCQs with Explanations
Day 3	**Deep Learning on HPC & Slurm Schedulers** • GEMM Matrix Computations in Neural Nets • Intel AMX Systolic Matrix Tile Engines • Slurm Workload Manager Architecture • Horovod Ring-AllReduce vs Parameter Server • Intel-Optimized TensorFlow & IPEX	`Day_03_Deep_Learning_HPC_Slurm_LMS.pptx`	`Day-03 Master Dossier`	`Day-03 live_demo/` `student_lab_practice/`	`Day-03_MCQs.md` 25 MCQs with Explanations
Day 4	**Distributed AI & Intel VTune Profiling** • Top-Down Microarchitecture Analysis (TMAM) • Retiring vs Bad Speculation vs Memory-Bound • Cache-Aware Roofline Model ($I = \text{FLOPs/Byte}$) • Dual-Socket NUMA Latency & `numactl` • InfiniBand HDR vs RoCE v2 Networking	`Day_04_Distributed_AI_VTune_Profiling_LMS.pptx`	`Day-04 Master Dossier`	`Day-04 live_demo/` `student_lab_practice/`	`Day-04_MCQs.md` 25 MCQs with Explanations

Day	Modules & Silicon Focus	Primary Slide Decks	Master Dossiers & Teaching Scripts	Hands-on Labs & Code Assets	Assessment & MCQs

Day 5	**Generative AI, LLMs on Intel HPC & OpenVINO**
	• Transformer Silicon Physics ($QK^T/\sqrt{d_k}$)
	• Prefill (Compute-Bound) vs Decode (Memory-Bound)
	• KV-Cache Sizing Formula & PagedAttention
	• Intel Gaudi 2/3 On-Die 24x 100GbE RoCE v2
	• OpenVINO NNCF INT8/INT4 Weight-Only Quantization
	• Distributed Fine-Tuning: IPEX, DeepSpeed, LoRA
	• FastAPI Production REST Inference Engine
		`Day_05_GenAI_LLMs_Gaudi_OpenVINO_LMS.pptx`
		`Day_05_Generative_AI_on_Intel_HPC.pptx`
			`Day-05 Master Dossier`
			`Day-05 Line-by-Line Teaching Guide`
			`Slide & Lab Synchronization Playbook`
				`Day_05_Master_Lab.ipynb`
				`live_demo/openvino_llm_benchmark.py`
				`live_demo/llm_fastapi_server.py`
					`Day-05_MCQs.md`
					25 MCQs with Explanations

🎯 3. Strategic Trainer Alignment (Shivam Gupta & Rajat Garg)

The Synchronized Teaching Architecture

As established in the pre-deployment sync call:

1. **Day 1 Division of Labor**:
   - **Rajat Garg (09:00 - 10:30)**: High-level Introduction to HPC, supercomputing taxonomy, industrial applications, Top500 supercomputers, and GLA LMS / attendance onboarding.
   - **Shivam Gupta (10:30 - 12:30)**: Intel Xeon silicon microarchitecture, memory wall, AVX-512 vector registers, Linux kernel architecture, automated shell scripting, passwordless SSH keys, and introduction to the Intel oneAPI ecosystem.
   - **Shivam Gupta (01:30 - 03:30)**: Intel oneAPI Base Toolkit vs HPC Toolkit, DPC++/SYCL single-source heterogeneous programming, Unified Shared Memory (`sycl::malloc_shared`), compilation pipelines (`icpx -fsycl`), Intel MPI, and live execution.
   - **Rajat Garg & Shivam Gupta (03:30 - 04:30)**: Supervised student hands-on lab execution in Google Colab / local Linux terminals, micro-activities, and Day 1 MCQ evaluation.
2. **Day 5 Division of Labor**:
   - **Shivam Gupta (09:00 - 12:30)**: Generative AI on HPC, transformer memory bandwidth bottleneck, Prefill vs Decode phases, KV-cache sizing math, Intel Xeon AMX vs Habana Gaudi 2/3 hardware, OpenVINO NNCF weight-only quantization (INT8/INT4), and distributed fine-tuning (IPEX, DeepSpeed, LoRA).
   - **Shivam Gupta & Rajat Garg (01:30 - 03:30)**: Hands-on OpenVINO LLM quantization lab, benchmarking tokens/second, deploying local FastAPI microservice, and capstone project execution.
   - **Rajat Garg (03:30 - 04:30)**: Enterprise GenAI economics, Capstone evaluation rubric, course valedictory, and certificate distribution.

💻 4. The 3-Tier Lab Execution Architecture (Colab vs Cluster vs Local)

To resolve the senior trainer's question ("*Colab me dikhaoge ya cluster pe ya GPU pe?*"), the lab courseware is architected across three foolproof tiers:

Execution Tier	Platform	Best For	Prerequisites	Status
Tier 1 (Universal Default)	**Google Colab (Interactive Notebooks)**	Zero-install, instant classroom delivery. Every student has a browser. Runs Intel DPC++/SYCL, C++ SIMD, OpenMP, and OpenVINO with pre-configured bash install cells.	Google Account	✅ Ready (`.ipynb` built with zero placeholders)
Tier 2 (Enterprise Hardware)	**Intel Developer Cloud (Free Cloud Cluster)**	Accessing real bare-metal Intel Data Center GPU Max 1100/1550, 4th/5th Gen Xeon Scalable CPUs with AMX, and Gaudi nodes. JupyterLab interface.	Intel Developer Cloud account	✅ Step-by-step connection guide included

Execution Tier	Platform	Best For	Prerequisites	Status
Tier 3 (Zero-Internet Safeguard)	**Local Linux Lab PCs / Pen Drive Dump**	Standalone deployment if GLA campus Wi-Fi throttles or fails. Fully offline pre-compiled wheels, datasets, and scripts.	USB 3.2 Pen Drive	✅ Pre-configured in `OFFLINE_PENDRIVE_TOOLKIT/`

🔒 5. Verification & Compliance Checklist

- 🟩 All 5 days have matching PowerPoint slide decks (both LMS 16:9 standard and NotebookLM high-density dark mode).
- 🟩 All 125 MCQs are validated with technical explanations and answer keys.
- 🟩 All Python and Bash scripts pass compilation and linting tests.
- 🟩 Comprehensive line-by-line teaching scripts and student defense matrices are published for Day 1 and Day 5.
- 🟩 Automated dry-run rehearsal script is prepared for the 3:00 PM pre-deployment sync call.